

Tasks

1.

On a Cisco Layer 3 switch, create three VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support', VLAN 30 for 'Marketing') and assign at least one interface to each VLAN.

Ans:

Step 1: Open Cisco Packet Tracer, drag a Multi-layer Switch (such as the 3560 model) onto the workspace, and connect three PCs to it.

Step 2: Click on the Layer 3 Switch, go to the CLI tab, and create the three required corporate business VLANs by typing:

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Sales
```

```
Switch(config-vlan)# vlan 20
```

```
Switch(config-vlan)# name Support
```

```
Switch(config-vlan)# vlan 30
```

```
Switch(config-vlan)# name Marketing
```

```
Switch(config-vlan)# exit
```

Step 3: Assign port FastEthernet0/1 directly to the Sales VLAN 10 network group:

```
Switch(config)# interface FastEthernet0/1
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 10
```

```
Switch(config-if)# exit
```

Step 4: Assign port FastEthernet0/2 directly to the Support VLAN 20 network group:

```
Switch(config)# interface FastEthernet0/2
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 20
```

```
Switch(config-if)# exit
```

Step 5: Assign port FastEthernet0/3 directly to the Marketing VLAN 30 network group:

```
Switch(config)# interface FastEthernet0/3
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 30
```

```
Switch(config-if)# exit
```

2.

Configure Switch Virtual Interfaces (SVIs) for each VLAN you created by assigning them unique IP addresses (e.g., 192.168.10.1/24 for VLAN 10, etc.) and ensure each SVI is in 'up/up' state.

Hint: Use the 'interface vlan' command to create SVIs and 'no shutdown' to enable them.

Ans:

Step 1: Create the virtual SVI gateway interface for VLAN 10 and assign its unique management IP address:

```
Switch(config)# interface vlan 10
```

```
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
```

```
Switch(config-if)# no shutdown
```

```
Switch(config-if)# exit
```

Step 2: Create the virtual SVI gateway interface for VLAN 20 and assign its unique management IP address:

```
Switch(config)# interface vlan 20
```

```
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
```

```
Switch(config-if)# no shutdown
```

```
Switch(config-if)# exit
```

Step 3: Create the virtual SVI gateway interface for VLAN 30 and assign its unique management IP address:

```
Switch(config)# interface vlan 30
```

```
Switch(config-if)# ip address 192.168.30.1 255.255.255.0
```

```
Switch(config-if)# no shutdown
```

```
Switch(config-if)# exit
```

Step 4: Return to the privileged EXEC setup mode line status check by typing end.

Step 5: Verify that all your configured SVI ports are running properly in an 'up/up' active state by executing:

```
Switch# show ip interface brief
```

3.

Enable IP routing on the Layer 3 switch so that devices in different VLANs can communicate with each other without using an external router.

Hint: Use the 'ip routing' global configuration command.

Step 1: Return to the primary device configuration terminal by typing configure terminal.

Step 2: Enter the mandatory global command to turn on the switch's internal hardware routing processor engine:

```
Switch(config)# ip routing
```

```
Switch(config)# exit
```

Step 3: By entering this command, the hardware transforms from a basic local network bridge into an active router capable of moving data packets across subnets without needing an external router.

4.

Verify inter-VLAN routing by connecting two PCs (one in VLAN 10 and one in VLAN 20) and pinging between them. Capture and submit the ping results to show successful communication.

Ans:

Step 1: Assign end-device IP parameters to your end host computers connected in Packet Tracer.

Step 2: Configure PC-Sales (VLAN 10) with IP 192.168.10.10 and Gateway 192.168.10.1. Configure PC-Support (VLAN 20) with IP 192.168.20.10 and Gateway 192.168.20.1.

Step 3: Click on PC-Sales, navigate to its Desktop app window, and open the command prompt.

Step 4: Type ping 192.168.20.10 and hit enter to test the internal hardware data switching path.

Step 5: Observe the successful CLI terminal output feedback confirming end-to-end data communication:

Pinging 192.168.20.10 with 32 bytes of data:

```
Reply from 192.168.20.10: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.20.10: bytes=32 time<1ms TTL=128
```

5.

Refactor your VLAN and SVI configuration so that VLAN 30 uses a different subnet mask (e.g., /25 instead of /24). Explain what changes you made and how this affects communication with the other VLANs.

Ans:

Step 1: Open the Layer 3 Switch configuration terminal to update the subnet masking settings for VLAN 30:

```
Switch# configure terminal
```

```
Switch(config)# interface vlan 30
```

```
Switch(config-if)# ip address 192.168.30.1 255.255.255.128
```

```
Switch(config-if)# exit
```

Step 2: Update the configuration on any PC residing inside the Marketing VLAN 30 network to match this modified subnet limit. Change its subnet mask from 255.255.255.0 to 255.255.255.128.

The subnet mask was modified from a standard Class C /24 network down to a sub-divided /25 variable length subnet mask, restricting the maximum available host range of that specific pool to 126 usable IP addresses (from .1 to .126).

This has **no negative effect** on cross-VLAN communication. Because the Layer 3 switch has ip routing active, it dynamically updates its internal routing table to recognize that VLAN 30 is now on a /25 network segment, allowing PCs in VLAN 10 and VLAN 20 to continue pinging devices in VLAN 30 perfectly.